

Lesson 2: Perimeter & Area

Friday, May 17, 2024 11:45 AM

Definition of Perimeter: A perimeter is a closed path that encompasses, surrounds, or outlines either a two dimensional shape or a one-dimensional length. The perimeter of a circle or an ellipse is called its circumference.

Outline the Perimeter of the following Polygons:

Polygons



Triangle



Quadrilateral



Pentagon



Hexagon



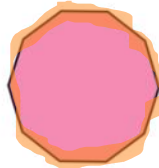
Heptagon



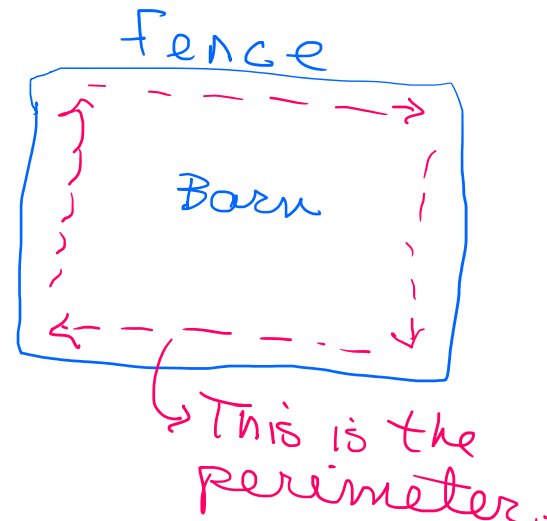
Octagon



Nonagon

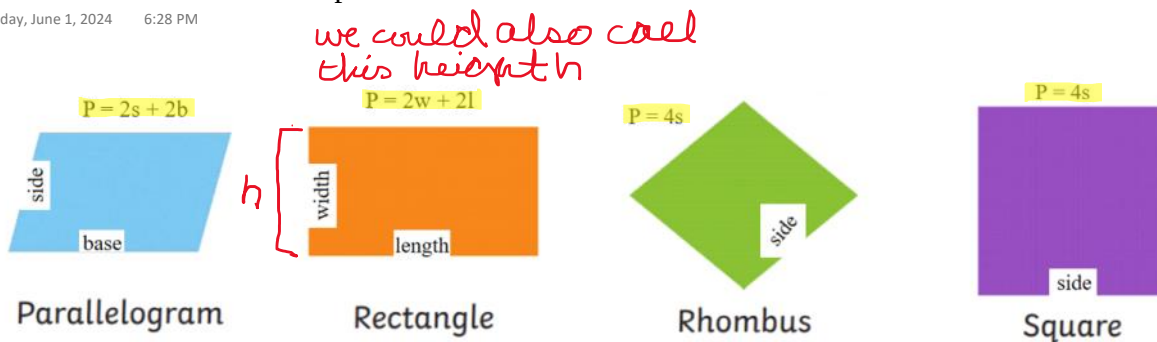


Decagon



Perimeter Formulas & Examples

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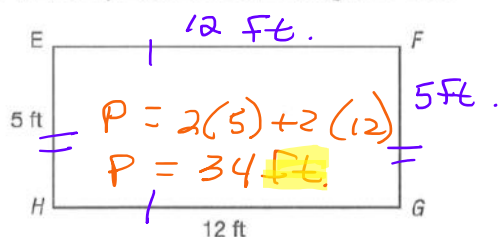


Many figures can be broken up into rectangles, triangles, and circles. Therefore, we do not need to memorize a lot of formulas. } Remember this !!

And, often, calculating Perimeter is simply a process of adding lengths of sides. easy!!

Solve.

1. Find the perimeter of rectangle $EFGH$.

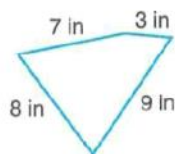


NOTICE: units are feet

2. One side of a square measures 6.5 centimeters. What is the perimeter of the square?

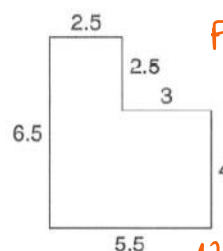
$$P = 4s = 4(6.5 \text{ cm}) = 26 \text{ cm.}$$

3. What is the distance around the figure?



$$P = 8 + 7 + 3 + 9 = 27 \text{ in.}$$

4. What is the perimeter of the six-sided figure?



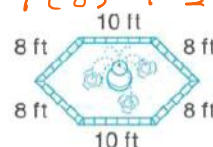
$$P = 6.5 + 2.5 + 2.5 + 3 + 4 + 5.5$$

$$P = 24$$

NOTICE: No units

5. The fountain in the diagram is enclosed by a low stone wall. What is the length of the wall?

$$P = 4(8) + 2(10) = 52 \text{ ft.}$$



ALWAYS STATE UNITS !!

FINDING UNKNOWN SIDES

Example The drawing shows a room from a dollhouse with measurements in inches. What is the perimeter of the room?

Before you can add all the sides, you need to find the missing sides, x and y .

Step 1

Look at the horizontal sides. The sum of the sides labeled 8 and x is equal to the side labeled 18.

$$8 + x = 18, \text{ so } x = 10$$

Step 2

Look at the vertical sides. The sum of the sides labeled 7 and 5 is equal to the side labeled y .

$$5 + 7 = y, \text{ so } y = 12$$

Step 3

Add the sides.

$$8 + 5 + 10 + 12 + 18 + 7 = 60$$

The perimeter of the room is **60 inches**.

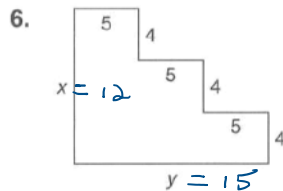
Because $8 + x = 18$

Because $7 + 5 = 12$

$$\therefore P = 7 + 8 + 5 + 10 + 12 + 18$$

$$P = 60$$

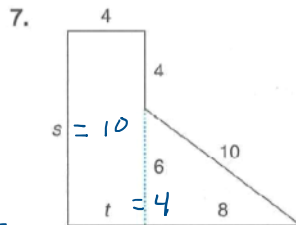
Find the perimeter of the following figures.



$$P = 12 + 3(5) + 3(4) + 15$$

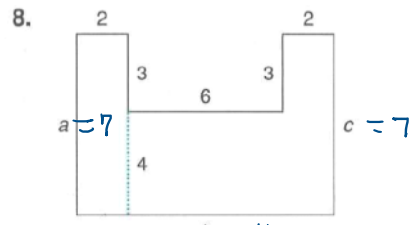
$$P = 54$$

Solve.



$$P = 10 + 4 + 4 + 10 + 8 + 4$$

$$P = 40$$

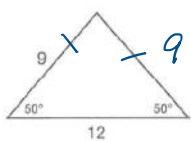


$$P = 7 + 2 + 3 + 6 + 3 + 2 + 7 + 10$$

$$P = 40$$

inches ???!!

9. Find the perimeter of the isosceles triangle.



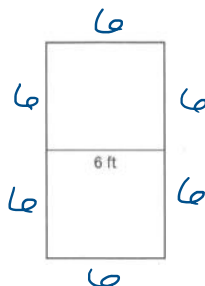
$$P = 2(9) + 12$$

$$P = 30$$

10. The rectangle is two stacked squares. What is the perimeter of the rectangle?

$$P = 6(6)$$

$$P = 36$$



11. **Use Algebra** A rectangular landscape is three times as long as it is wide. If the perimeter of the landscape is 64 inches, what is the length?

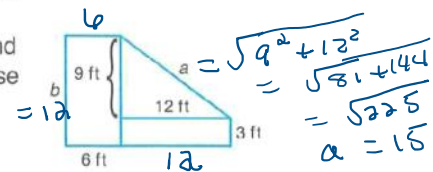
$$2(x) + 2(3x) = 64$$

$$8x = 64, x = 8, L = 3(8), \therefore \text{length} = 24 \text{ in.}$$

12. **Explain** The stage area shown consists of two rectangular platforms and a triangular one. Suppose you need to find the perimeter of the stage.

Explain how you could find the measures of sides a and b .

see diagram



$$a = \sqrt{9^2 + 12^2}$$

$$= \sqrt{81 + 144}$$

$$= \sqrt{225}$$

$$a = 15$$

Area

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Definition of Area: an area is the measure of a region's size on a surface.

Finding Area: the surface is divided into squares of equal size. The squares could be, for example, 1 inch by 1 inch, 1 foot by 1 foot, 1 acre by 1 acre, or 1 unit by 1 unit.

For Example: this surface is 4 by 4 = 16 square units.

you MUST state units!!

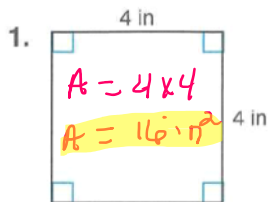
1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	16

Formula for Area of a Rectangle: $A = l \cdot w = \text{length} \cdot \text{width}$

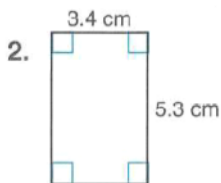
$A = \underline{\hspace{2cm}} \text{ sq. units}$

In the examples below, for #4 & #5, break up the figure.

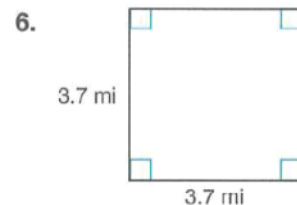
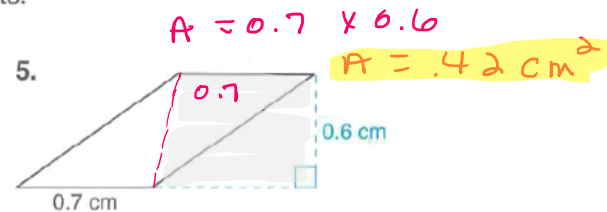
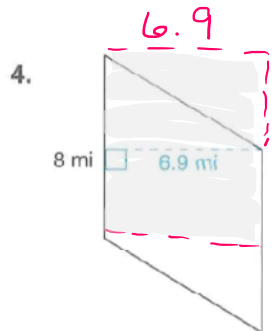
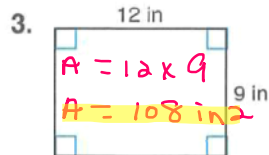
Find the area of the figures. Remember to label your answer in square units.



OR $A = 16 \text{ sq. in.}$



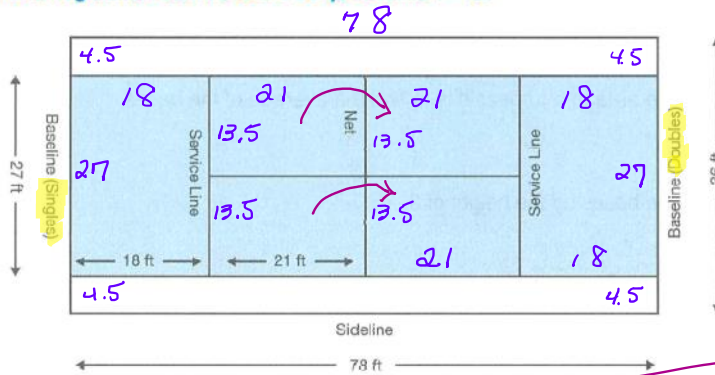
$A = 3.4 \times 5.3$
 $A = 18.02 \text{ cm}^2$



$A = 3.7^2$
 $A = 13.69 \text{ mi}^2$

$\therefore$ we may assume it is perfectly symmetrical.

Use the diagram of a regulation tennis court for problems 7–10.



Remember: proving FALSE requires only ONE counter-example.

Rectangle #1

7. A singles court will be painted a contrasting color. How many square feet will be painted?

$$A = 27 \times 78 = 2106 \text{ ft.}^2$$

8. On a singles court, what is the area of the rectangle between the service line and the baseline? $A = 27 \times 18 =$

$$A = 27 \times 18$$
$$A = 486 \text{ Ft}^2$$

9. A ball being served must land in the opposite small rectangle adjacent to the net. What is the area of the space where a serve must land?

$$A = 13.5 \times 21$$

$$A = 283.5 \text{ sq. ft.}$$

10. There are 9 square feet in 1 square yard. What is the area of a regulation doubles court in square yards? $A = (36 \times 78) / 9$

$$A = (36 \times 78) / 9$$

$$A = 312 \text{ sq. yds.}$$

14. **Explain** Two rectangles have different dimensions but the same area. Do the rectangles have the same perimeter? Explain your thinking. Draw an example to support your reasoning.

NO. FALSE.

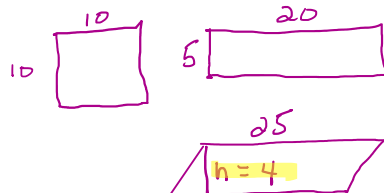
12. **Draw** A figure has an area of 100 square feet. Draw and label a square, a rectangle, and a parallelogram, each with an area of 100 square feet.

5 $\boxed{\begin{matrix} P = 18 \text{ units} \\ A = 20 \text{ units} \end{matrix}}$ 2

4

Rectangle #2

$P = 24 \text{ units}$
 $A = 20 \text{ units}$
 10



Be sure to label height,
not width!!!

Area of Triangles

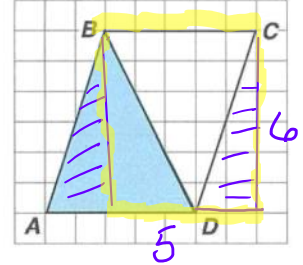
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This is a great explanation for the area of a triangle:

your instructor enjoys cutting apart a shape and putting it back together in an easier way!!!

Triangles are related to parallelograms. Study the drawing to the right. Compare triangle ABD to parallelogram $ABCD$. What do you notice about the area of the shapes? How are they related?

The area of the triangle is one-half the area of the parallelogram. If you know one, you can solve for the other. Parallelogram $ABCD$ has a base of 5 and a height of 6. Find the areas of both figures.



Area of $ABCD$:	base $\times$ height	$5 \times 6 = 30$ square units
Area of ABD :	base $\times$ height $\div 2$	$5 \times 6 \div 2 = 15$ square units

area of Parallelogram
 $A = 5 \times 6 = 30 \text{ units}^2$

area of triangle
 $A = \frac{1}{2}(30)$
 $= 15 \text{ units}^2$

You can write the following formula based on the example.

Area (A) of a Triangle

$$A = \frac{1}{2}bh, \text{ where } b = \text{base and } h = \text{height}$$

Area of Trapezoids

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Look at trapezoid $KLMN$. To find its area, you need to multiply base by height, but a trapezoid has two bases.

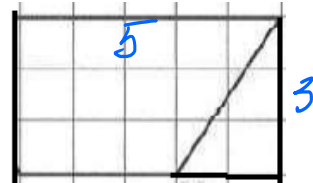
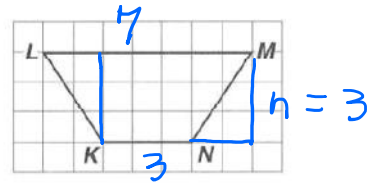
The first step in finding the area of a trapezoid is to find the average of the bases.

$$\frac{LM + KN}{2} = \frac{7 + 3}{2} = \frac{10}{2} = 5$$

Multiply the average of the bases by the height of 3.

$$5 \times 3 = 15 \text{ square units}$$

You are given the formula for finding the area of a trapezoid on the GED math formula sheet.

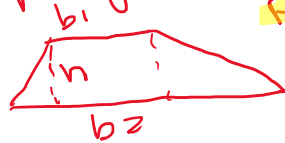


Area (A) of a Trapezoid

$$A = \frac{1}{2}h(b_1 + b_2), \text{ where } h = \text{height and } b_1 \text{ and } b_2 \text{ are the bases}$$

Note: in the above diagram, the trapezoid is horizontally symmetrical, but not all trapezoids are!!

E.g.:

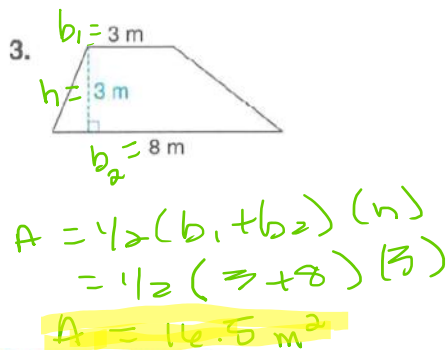
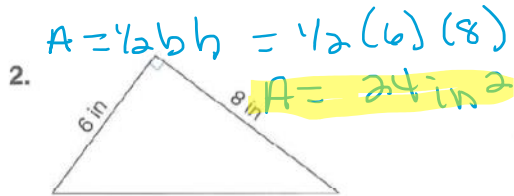
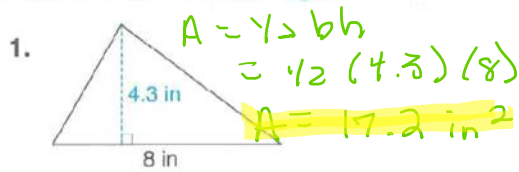


Formula still applies.

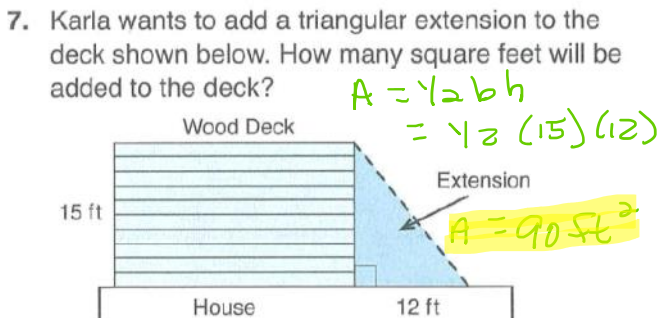
Triangles & Trapezoids - Example

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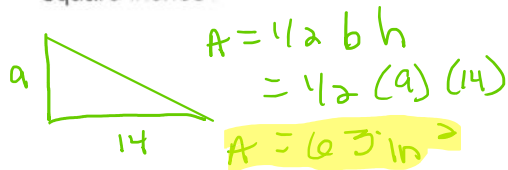
Find the area of each figure.



Solve as directed.



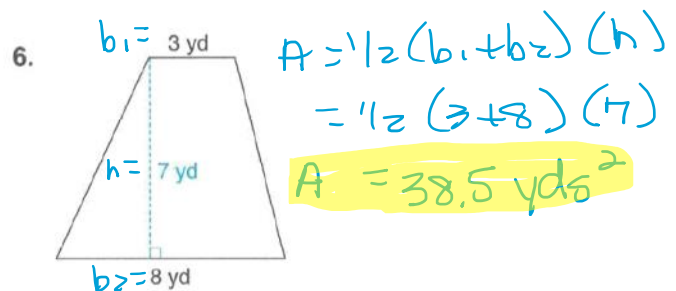
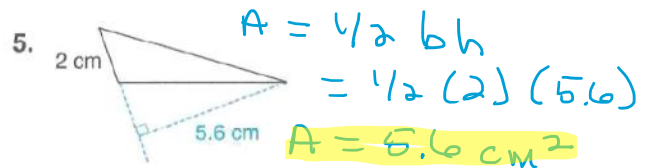
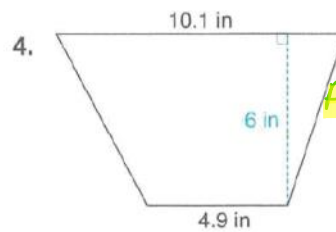
8. The two legs of a right triangle are 9 inches and 14 inches. What is the area of the triangle in square inches?



$$A = \frac{1}{2}(b_1 + b_2)h$$

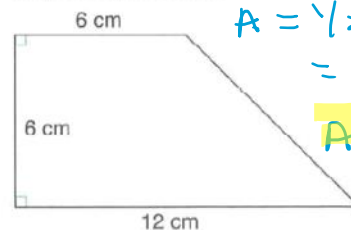
$$= \frac{1}{2}(10.1 + 4.9)(6)$$

$$A = 45 \text{ in}^2$$



9. **Explore** One way to find the area of a figure is to cut it into smaller shapes.

- a. Find the area of the trapezoid using the trapezoid formula.

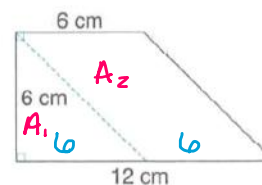


$$A = \frac{1}{2}(b_1 + b_2)(h)$$

$$= \frac{1}{2}(6 + 12)(6)$$

$$A = 54 \text{ cm}^2$$

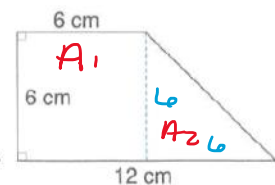
- b. The trapezoid has been divided into smaller shapes in two different ways. Find the area of the individual pieces and add to find the total area.



$$A_1 = \frac{1}{2}bh$$

$$= \frac{1}{2}(6)(6)$$

$$A_1 = 18 \text{ cm}^2$$



$$A_1 = 6^2$$

$$= 6(6)$$

$$A_1 = 36 \text{ cm}^2$$

$$A_2 = bh$$

$$= 4(6)$$

$$A_2 = 36 \text{ cm}^2$$

$$A = A_1 + A_2$$

$$A = 54 \text{ cm}^2$$

$$A_2 = \frac{1}{2}bh$$

$$= \frac{1}{2}(6)(6)$$

$$A_2 = 18 \text{ cm}^2$$

$$A = A_1 + A_2$$

$$A = 54 \text{ cm}^2$$